

NIKOLAI, AK, USA

WMO: 702625

Lat: 63.017N

Lon: 154.367W

Elev: 414

StdP: 14.48

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00104

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.7	-37.8	-34.9	0.8	-27.7	-32.7	0.9	-26.9	22.7	8.5	20.3	10.3	0.2	270	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	79.5	60.4	74.8	58.1	71.7	56.5	62.1	74.9	60.1	70.9	58.5	68.0	6.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.3	71.2	63.8	55.4	66.4	62.4	54.2	63.4	61.7	27.9	74.4	26.6	70.5	25.4	68.2	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.4	13.9	DB	-45.8	85.3	7.2	4.2	-51.0	88.3	-55.2	90.7	-59.3	93.1	-64.5	96.1	
			WB	-30.8	63.8	7.1	3.1	-35.9	66.1	-40.0	67.9	-44.0	69.6	-49.2	71.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.3	-2.4	7.2	14.6	32.6	47.6	56.6	59.6	54.3	45.0	29.0	5.8	0.3
	DBStd	25.59	20.59	15.07	13.78	10.45	7.60	5.35	5.52	5.51	7.00	11.15	16.22	17.84
	HDD50	8292	1624	1198	1099	521	132	5	1	18	173	653	1327	1541
	HDD65	13056	2089	1618	1564	971	540	258	183	333	600	1117	1777	2006
	CDD50	737	0	0	0	0	57	204	300	152	23	1	0	0
	CDD65	27	0	0	0	0	0	8	17	2	0	0	0	0
	CDH74	435	0	0	0	0	14	149	240	32	0	0	0	0
	CDH80	91	0	0	0	0	1	42	44	4	0	0	0	0
	WSAvg	4.8	5.0	5.3	5.6	5.9	5.1	4.5	4.2	3.9	4.3	4.6	4.2	4.3
Precipitation	PrecAvg	23.3	1.6	1.4	1.1	0.9	1.2	2.0	2.8	3.3	2.9	2.2	1.9	1.9
	PrecMax	29.3	3.0	3.9	2.7	1.8	2.2	3.7	5.6	5.2	5.7	6.5	5.2	3.9
	PrecMin	17.5	0.2	0.1	0.1	0.2	0.4	0.8	1.5	1.1	1.1	0.5	0.2	0.4
	PrecStd	3.1	0.8	0.7	0.6	0.4	0.5	0.7	0.9	0.9	1.4	1.2	1.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	39.2	47.7	61.3	77.4	85.8	85.0	78.4	67.8	56.8	43.0	41.3
		MCWB	34.1	32.5	37.9	45.0	56.1	59.9	63.8	61.4	53.9	45.6	40.3	33.5
	2%	DB	35.2	34.9	41.1	56.1	72.2	78.5	80.7	72.7	63.6	50.2	36.0	33.0
		MCWB	31.6	31.2	34.5	42.1	53.5	59.0	61.2	58.6	51.7	43.0	33.7	30.6
	5%	DB	29.7	31.8	37.1	53.0	67.0	73.1	76.2	68.4	60.3	45.8	31.8	27.4
		MCWB	27.5	29.4	31.4	41.0	51.2	55.8	60.4	56.8	50.5	40.9	30.1	25.5
	10%	DB	22.9	27.3	33.6	48.3	63.0	70.0	72.3	64.5	56.0	42.7	27.0	23.1
		MCWB	21.0	24.7	30.0	38.6	48.5	54.7	58.5	54.8	49.0	39.4	25.3	21.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.3	34.7	38.4	45.9	56.8	63.3	66.1	62.7	55.1	46.8	39.4	35.3
		MCDB	40.4	38.2	45.5	60.1	74.5	78.6	79.8	74.5	63.4	53.9	44.4	40.4
	2%	WB	32.2	31.7	35.1	43.0	54.1	60.0	63.1	60.2	53.2	43.8	33.8	30.9
		MCDB	35.8	34.5	40.3	54.3	70.1	73.5	77.9	68.6	60.7	48.9	35.7	33.8
	5%	WB	28.3	29.0	32.4	41.2	51.7	57.8	61.3	58.3	51.4	41.5	29.8	26.4
		MCDB	30.6	32.0	36.4	51.7	65.0	69.8	73.1	65.3	57.4	45.5	31.9	27.8
	10%	WB	22.9	25.1	29.6	39.1	49.8	55.7	59.5	56.7	49.9	39.3	25.4	22.7
		MCDB	25.0	27.4	33.8	48.2	61.0	66.8	69.8	63.0	55.9	42.8	27.2	24.0
Mean Daily Temperature Range	5% DB	MDBR	14.3	16.9	20.9	20.0	21.3	19.5	18.0	16.1	15.4	11.6	13.6	13.7
		MCDBR	14.8	16.8	21.2	25.1	29.0	26.6	26.1	22.6	22.9	15.4	12.9	16.9
		MCWBR	12.0	14.2	15.5	14.2	13.9	10.9	11.1	10.4	13.1	10.1	10.9	14.7
	5% WB	MCDBR	15.0	15.8	17.4	23.5	25.6	23.1	23.8	17.7	18.6	13.6	12.9	16.4
		MCWBR	12.2	13.6	12.7	13.4	12.5	10.3	10.7	8.6	11.4	9.0	11.0	14.2
Clear-Sky Solar Irradiance	taub		0.199	0.256	0.272	0.323	0.362	0.377	0.377	0.382	0.310	0.282	0.213	0.191
	taud		2.024	2.231	2.272	2.245	2.365	2.341	2.385	2.377	2.558	2.454	2.022	1.911
	Ebn at Noon		145	223	266	272	271	267	264	251	255	216	130	71
	Edh at Noon		10	19	26	34	33	35	33	30	21	15	10	5
All-Sky Solar Radiation	RadAvg		79	307	880	1459	1638	1685	1445	1102	726	311	104	33
	RadStd		10	30	79	108	156	113	126	111	101	38	17	3

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+1.80	N/A	+1.53	N/A	N/A	N/A	-538	-644	N/A	N/A

Nomenclature: See separate page